



Experiential Learning - Phase 1

IVth Semester

Soccer Player Injury Prediction

An XGBoost-Based Binary Classification Model for Predicting Major Injuries in Premier League Players

Snehal Reddy Thadigotla(1RV24IS126)

Sumukha Upadhyaya(1RV24IS130)

Vikas Y K (1RV24CI139)



Introduction

The Injury Problem: Soccer player injuries are among the costliest and least predictable challenges in professional sport, with roughly 20% of scheduled matches missed to injury each season.

The Traditional Flaw: Injury management relies on physiotherapist intuition and generic load thresholds, with no objective, player-specific risk assessment before squad selection.

Our Approach: We apply machine learning to a player's physical attributes, injury history, and playing load to quantify the risk of a major injury before it occurs.

The Actionable Outcome: An XGBoost classifier predicts whether a Premier League player will suffer a major injury (over 120 days out) in the upcoming season, informing coaching decisions.

Problem Statement

Intuition over Data: Coaching staff select potentially injury-prone players based on feel rather than a quantified, data-driven risk score, leading to avoidable re-injuries.

Fragmented Data Sources: Player attributes, injury history, and game-time statistics live in three separate, heterogeneous datasets with no common player identifier, making joint analysis difficult.

Severe Class Imbalance: Only about 25% of player-seasons end in a major injury, so a naive classifier biases heavily toward the majority non-injury class.

Reactive, Not Predictive: Clubs analyse injury data after the fact instead of forecasting risk before games, missing the chance to rest at-risk players proactively.

Primary Objectives

- **Build a Binary Injury Classifier:** Train an XGBoost model to predict whether a player will sustain a major injury (over 120 days out) in the upcoming season.
- **Integrate Multi-Source Data:** Merge FIFA 16-21 player attributes, Transfermarkt injury history, and Barclays Premier League game-time statistics into a unified 685-row dataset.
- **Engineer Predictive Features:** Derive cumulative minutes played, average days injured in prior seasons, significant injury flags, and cumulative days injured.
- **Benchmark Performance:** Evaluate the model on a 30% holdout test set using Accuracy, Precision, Recall, F1-Score, and AUC-ROC.

Secondary Objectives

- **Handle Class Imbalance:** Apply SMOTE oversampling to grow the minority class in the training set from 479 to 734 synthetic samples, reducing model bias.
- **Optimise Hyperparameters:** Run GridSearchCV across XGBoost parameters to find the optimal configuration: 200 trees, max depth 4, learning rate 0.1, subsample 0.8-0.9.
- **Interpret the Model:** Generate feature importance rankings to reveal which attributes most strongly drive injury predictions, giving coaches actionable insight.
- **Define Responsible Use:** Establish ethical boundaries so model outputs inform selection discussions only, and are never used to discriminate against players.

Literature Review

- **Chen & Guestrin (2016), "XGBoost: A Scalable Tree Boosting System" (ACM KDD):** Introduced the regularised gradient-boosting framework used in this project, achieving state-of-the-art results. Limitation: requires careful hyperparameter tuning and can overfit small datasets.
- **Chawla et al. (2002), "SMOTE: Synthetic Minority Over-sampling Technique" (JAIR):** Proposed synthetic minority oversampling via k-NN interpolation to address class imbalance, the technique used here. Limitation: may introduce noise where class boundaries overlap.
- **Claudino et al. (2019), "AI for Injury Risk Assessment in Team Sports" (BMJ Open Sport & Exercise Medicine):** Systematic review finding ML models outperform traditional statistics for sports injury prediction. Limitation: high variability in features and data quality across studies.

Literature Review

- **Rossi et al. (2018), "Effective Injury Forecasting in Soccer with GPS Training Data and ML" (PLOS ONE):** First to forecast soccer injuries from GPS load data using Random Forest at 85% accuracy. Limitation: requires costly GPS hardware unavailable for historical data.
- **Gabbett (2016), "The Training-Injury Prevention Paradox" (British Journal of Sports Medicine):** Showed that high training load both causes and prevents injury, introducing the acute:chronic workload ratio (ACWR). Limitation: ACWR alone is insufficient for individual-level prediction.
- **Pedregosa et al. (2011), "Scikit-learn: Machine Learning in Python" (JMLR):** Introduced the standard Python ML library providing GridSearchCV and evaluation tools used here. Limitation: XGBoost and SMOTE require separate add-on packages.

Methodology

Step 1: Data Collection

- FIFA 16-21 attributes and Barclays Premier League stats were downloaded from Kaggle; Transfermarkt injury histories were scraped using the worldfootballR R package.

Step 2: Data Merging & Feature Engineering

- The three sources were joined on player and season into 685 records across 317 players (2016/17-2020/21). Cumulative features were derived: minutes played, average days injured in prior seasons, and significant-injury flags.

Step 3: Class Balancing

- SMOTE oversampling was applied to the 70% training split, expanding it from 479 to 734 balanced samples to counter the 25:75 class imbalance.

Methodology

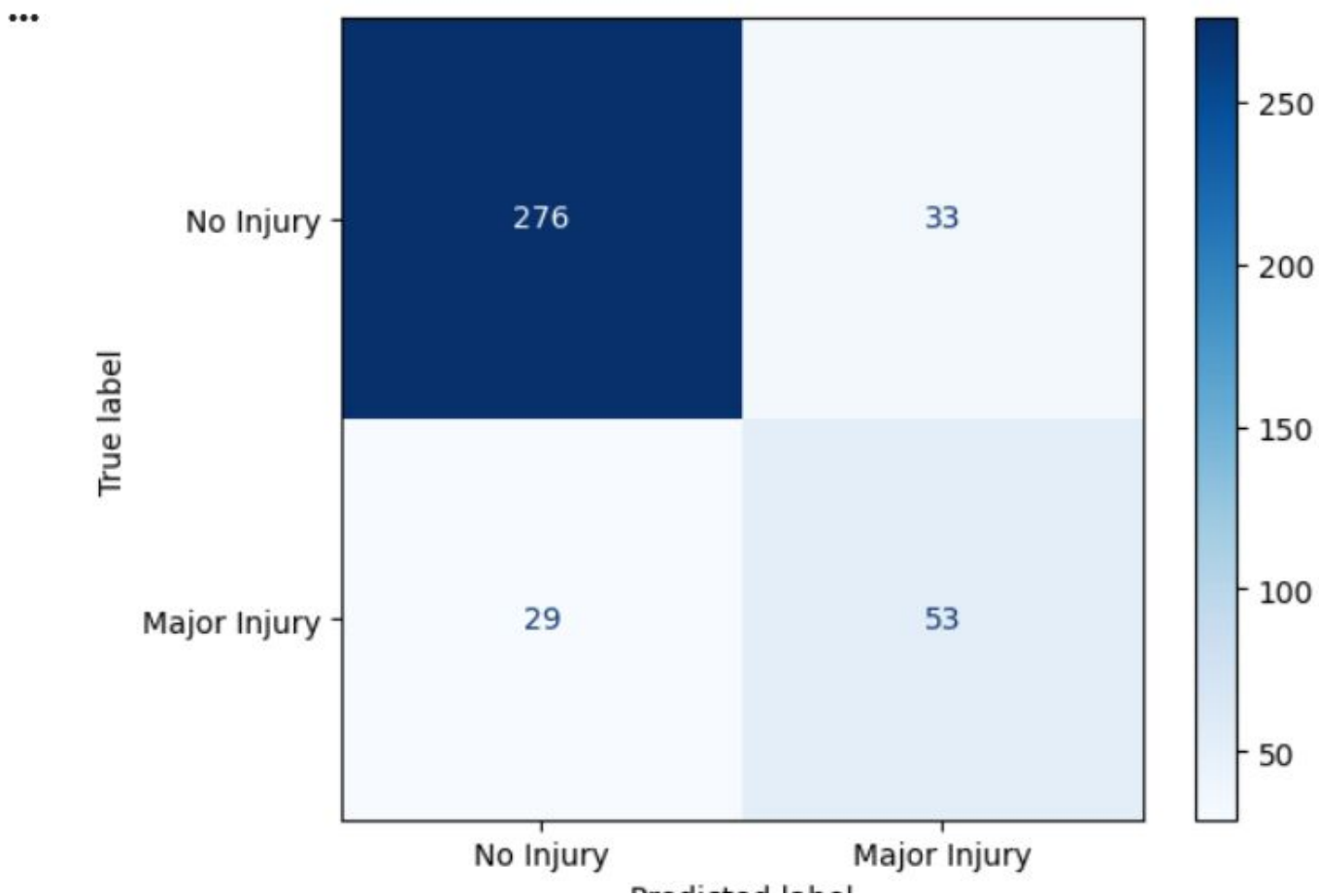
Step 4: Model Training

- XGBoost was trained with GridSearchCV over n_estimators, max_depth, learning_rate, min_child_weight and subsample. Optimal configuration: 200 trees, depth 4, learning rate 0.1, subsample 0.8-0.9.

Step 5: Evaluation

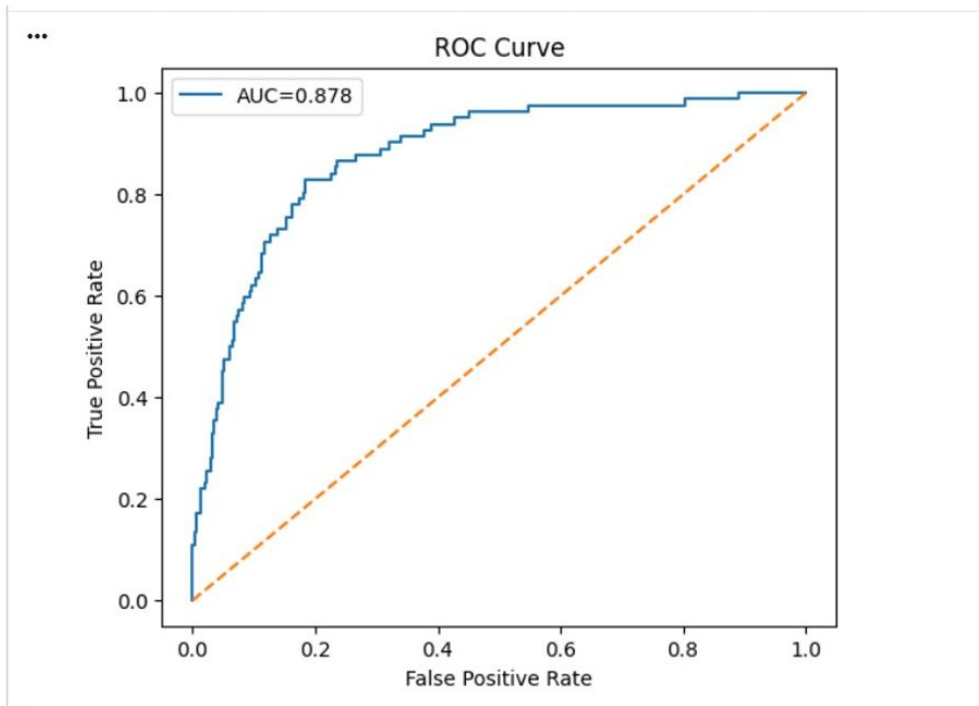
- Model performance on the 206-sample holdout test set:
 - **Accuracy: 84.14% | AUC-ROC: 87.84%**
 - **Precision: 61.63% | Recall: 64.63% | F1-Score: 63.1%**
- Feature importance identified prior injury history and player age as the strongest predictors of a major injury.

Results



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Precision: 61.63%
Recall: 64.63%
F1-Score: 63.1%

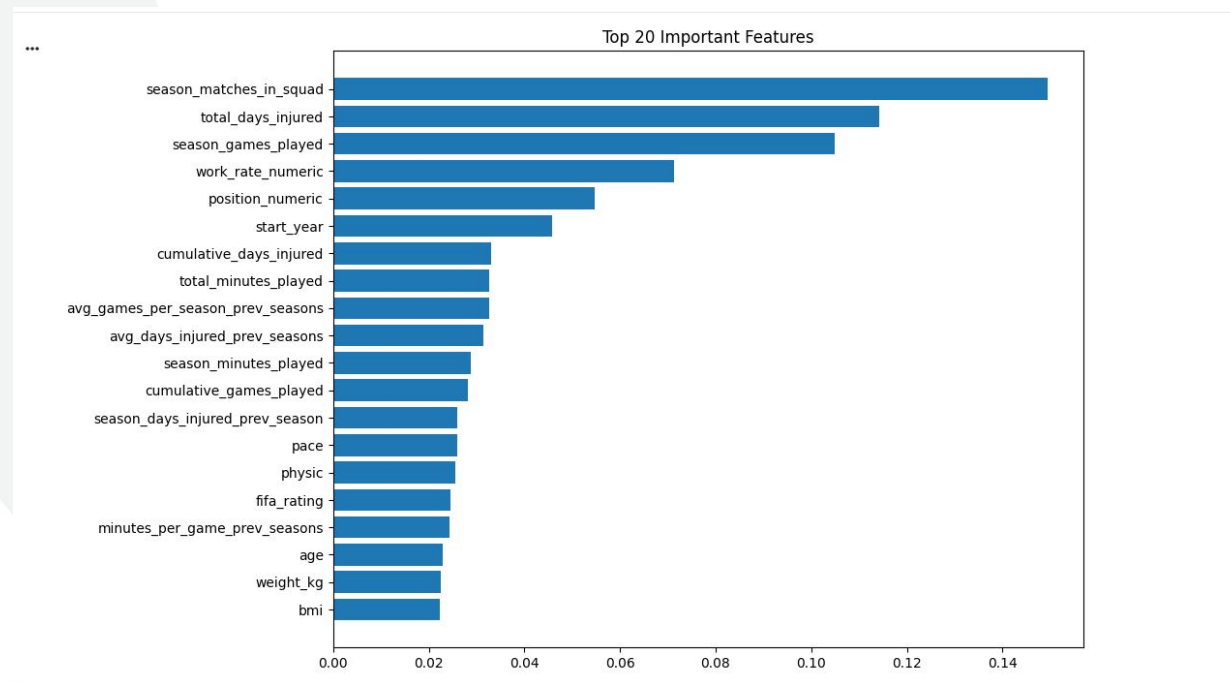
Results



Area Under ROC Curve

Measures:

How well the model separates: Injured Players vs Non-Injured Players



Feature importance graph shows which player attributes had the greatest influence on the XGBoost model's injury predictions, helping us identify the key factors associated with major injuries.

Tools and Technologies Used

- **Data Collection & Processing:**
 - R with worldfootballR (Transfermarkt scraping); Kaggle API for datasets
 - Python 3, Pandas & NumPy (data parsing, merging, normalisation)
- **Machine Learning & Evaluation:**
 - XGBoost (gradient boosted tree classifier)
 - Scikit-learn (GridSearchCV, train/test split, metrics)
 - imbalanced-learn (SMOTE oversampling)
- **Visualisation & Development:**
 - Matplotlib & Seaborn (feature importance, confusion matrix)
 - Jupyter Notebook (development); Git & GitHub (version control)

Mapping to IVth Semester Courses

Design and Analysis of Algorithms

- XGBoost is an ensemble of greedy best-split decision trees - a direct application of tree algorithm design.
- Grid search is exhaustive combinatorial search over a discrete hyperparameter space (brute-force search design).
- SMOTE uses k-nearest-neighbour interpolation, applying nearest-neighbour graph concepts; $O(n.k)$ per sample.

Computer Networks

- The Kaggle API and worldfootballR operate over HTTP REST - the client-server request/response model.
- Web scraping involves TCP connections, HTTP GET requests, and response parsing across the protocol stack.

Mapping to IVth Semester Courses

Internet of Things and Applications

- Edge AI: the trained model can run on edge devices (Raspberry Pi, Jetson Nano) for real-time injury risk scoring at training grounds.
- Wearable sensors: GPS, accelerometer and heart-rate data from player wearables are the live-stream equivalent of this project's load features.
- Sensor fusion: combining three heterogeneous data sources mirrors fusing multiple IoT sensor streams into one unified data layer.
- Real-time responsiveness: low-latency risk updates at selection time parallel the time-critical sensing constraints of IoT systems.

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